He[g] HELIUM (GAS) 4.003

Phase	T [K]	C_p	S	−(G−H298)/T	H	H−H298	G	ΔH_f	ΔG_f	log K_f
		[J / (K mol)		]	[kJ / mol				]	[−]
GAS	298.15	20.786	126.148	126.148	0.000	0.000	−37.611	0.000	0.000	0.000
	300.00	20.786	126.277	126.149	0.038	0.038	−37.845	0.000	0.000	0.000
	400.00	20.786	132.257	126.964	2.117	2.117	−50.786	0.000	0.000	0.000
	500.00	20.786	136.895	128.504	4.196	4.196	−64.252	0.000	0.000	0.000
	600.00	20.786	140.685	130.228	6.274	6.274	−78.137	0.000	0.000	0.000
	700.00	20.786	143.889	131.956	8.353	8.353	−92.369	0.000	0.000	0.000
	800.00	20.786	146.665	133.625	10.432	10.432	−106.900	0.000	0.000	0.000
	900.00	20.786	149.113	135.213	12.510	12.510	−121.691	0.000	0.000	0.000
	1000.00	20.786	151.303	136.714	14.589	14.589	−136.714	0.000	0.000	0.000
	1100.00	20.786	153.284	138.132	16.667	16.667	−151.945	0.000	0.000	0.000
	1200.00	20.786	155.093	139.471	18.746	18.746	−167.365	0.000	0.000	0.000
	1300.00	20.786	156.756	140.737	20.825	20.825	−182.959	0.000	0.000	0.000
	1400.00	20.786	158.297	141.937	22.903	22.903	−198.712	0.000	0.000	0.000
	1500.00	20.786	159.731	143.076	24.982	24.982	−214.614	0.000	0.000	0.000
	1600.00	20.786	161.072	144.160	27.060	27.060	−230.655	0.000	0.000	0.000
	1700.00	20.786	162.332	145.192	29.139	29.139	−246.826	0.000	0.000	0.000
	1800.00	20.786	163.521	146.177	31.218	31.218	−263.119	0.000	0.000	0.000
	1900.00	20.786	164.644	147.120	33.296	33.296	−279.528	0.000	0.000	0.000
	2000.00	20.786	165.711	148.023	35.375	35.375	−296.046	0.000	0.000	0.000
	2100.00	20.786	166.725	148.890	37.453	37.453	−312.669	0.000	0.000	0.000
	2200.00	20.786	167.692	149.723	39.532	39.532	−329.390	0.000	0.000	0.000
	2300.00	20.786	168.616	150.524	41.611	41.611	−346.206	0.000	0.000	0.000
	2400.00	20.786	169.500	151.297	43.689	43.689	−363.112	0.000	0.000	0.000
	2500.00	20.786	170.349	152.042	45.768	45.768	−380.104	0.000	0.000	0.000
	2600.00	20.786	171.164	152.762	47.847	47.847	−397.180	0.000	0.000	0.000
	2700.00	20.786	171.949	153.458	49.925	49.925	−414.336	0.000	0.000	0.000
	2800.00	20.786	172.705	154.132	52.004	52.004	−431.569	0.000	0.000	0.000
	2900.00	20.786	173.434	154.785	54.082	54.082	−448.876	0.000	0.000	0.000
	3000.00	20.786	174.139	155.418	56.161	56.161	−466.255	0.000	0.000	0.000

References

Phase	H / S	C_p
GAS	Hu1	Hu1

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178.490 HAFNIUM **Hf**

Phase	T [K]	C_p	S	−(G−H298)/T	H	H−H298	G	ΔH_f	ΔG_f	log K_f
		[J / (K mol)		]	[kJ / mol				]	[−]
SOL−A	298.15	25.736	43.555	43.555	0.000	0.000	−12.986	0.000	0.000	0.000
	300.00	25.750	43.715	43.556	0.048	0.048	−13.067	0.000	0.000	0.000
	400.00	26.512	51.227	44.575	2.661	2.661	−17.830	0.000	0.000	0.000
	500.00	27.274	57.225	46.525	5.350	5.350	−23.262	0.000	0.000	0.000
	600.00	28.036	62.265	48.739	8.116	8.116	−29.243	0.000	0.000	0.000
	700.00	28.798	66.644	50.991	10.957	10.957	−35.693	0.000	0.000	0.000
	800.00	29.560	70.539	53.195	13.875	13.875	−42.556	0.000	0.000	0.000
	900.00	30.321	74.064	55.321	16.869	16.869	−49.789	0.000	0.000	0.000
	1000.00	31.083	77.299	57.359	19.939	19.939	−57.359	0.000	0.000	0.000
	1100.00	31.845	80.297	59.310	23.086	23.086	−65.241	0.000	0.000	0.000
	1200.00	32.607	83.100	61.177	26.308	26.308	−73.412	0.000	0.000	0.000
	1300.00	33.369	85.740	62.966	29.607	29.607	−81.855	0.000	0.000	0.000
	1400.00	34.131	88.241	64.683	32.982	32.982	−90.556	0.000	0.000	0.000
	1500.00	34.892	90.622	66.333	36.433	36.433	−99.500	0.000	0.000	0.000
	1600.00	35.654	92.898	67.923	39.961	39.961	−108.676	0.000	0.000	0.000
	1700.00	36.416	95.083	69.457	43.564	43.564	−118.076	0.000	0.000	0.000
	1800.00	37.178	97.186	70.939	47.244	47.244	−127.690	0.000	0.000	0.000
	1900.00	37.940	99.216	72.374	51.000	51.000	−137.511	0.000	0.000	0.000
	2000.00	38.702	101.182	73.766	54.832	54.832	−147.531	0.000	0.000	0.000
	2013.00	38.801	101.433	73.944	55.336	55.336	−148.848	0.000	0.000	0.000
			3.346		6.736					
SOL−B	2013.00	36.819	104.779	73.944	62.072	62.072	−148.848	0.000	0.000	0.000
	2100.00	36.819	106.337	75.253	65.275	65.275	−158.032	0.000	0.000	0.000
	2200.00	36.819	108.050	76.706	68.957	68.957	−168.752	0.000	0.000	0.000
	2300.00	36.819	109.686	78.104	72.639	72.639	−179.640	0.000	0.000	0.000
	2400.00	36.819	111.253	79.453	76.321	76.321	−190.687	0.000	0.000	0.000
	2500.00	36.819	112.756	80.755	80.003	80.003	−201.888	0.000	0.000	0.000
			9.623		24.058					
LIQ	2500.00	33.472	122.380	80.755	104.061	104.061	−201.888	0.000	0.000	0.000
	2600.00	33.472	123.692	82.382	107.408	107.408	−214.192	0.000	0.000	0.000
	2700.00	33.472	124.956	83.935	110.755	110.755	−226.625	0.000	0.000	0.000
	2800.00	33.472	126.173	85.422	114.102	114.102	−239.182	0.000	0.000	0.000
	2900.00	33.472	127.347	86.848	117.449	117.449	−251.858	0.000	0.000	0.000
	3000.00	33.472	128.482	88.217	120.797	120.797	−264.650	0.000	0.000	0.000
	3100.00	33.472	129.580	89.533	124.144	124.144	−277.553	0.000	0.000	0.000
	3200.00	33.472	130.642	90.801	127.491	127.491	−290.565	0.000	0.000	0.000
	3300.00	33.472	131.672	92.024	130.838	130.838	−303.681	0.000	0.000	0.000
	3400.00	33.472	132.672	93.205	134.185	134.185	−316.898	0.000	0.000	0.000
	3500.00	33.472	133.642	94.347	137.533	137.533	−330.214	0.000	0.000	0.000
	3600.00	33.472	134.585	95.452	140.880	140.880	−343.626	0.000	0.000	0.000
	3700.00	33.472	135.502	96.522	144.227	144.227	−357.130	0.000	0.000	0.000
	3800.00	33.472	136.395	97.559	147.574	147.574	−370.725	0.000	0.000	0.000
	3900.00	33.472	137.264	98.566	150.921	150.921	−384.408	0.000	0.000	0.000
	4000.00	33.472	138.111	99.544	154.269	154.269	−398.177	0.000	0.000	0.000

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Hf HAFNIUM [continued] 178.490

Phase	T [K]	C_p	S	−(G−H298)/T	H	H−H298	G	ΔH_f	ΔG_f	log K_f
		[J / (K mol)		]	[kJ / mol				]	[−]
LIQ	4100.00	33.472	138.938	100.495	157.616	157.616	−412.030	0.000	0.000	0.000
	4200.00	33.472	139.745	101.420	160.963	160.963	−425.964	0.000	0.000	0.000
	4300.00	33.472	140.532	102.321	164.310	164.310	−439.978	0.000	0.000	0.000
	4400.00	33.472	141.302	103.198	167.657	167.657	−454.070	0.000	0.000	0.000
	4500.00	33.472	142.054	104.053	171.005	171.005	−468.238	0.000	0.000	0.000
	4600.00	33.472	142.790	104.887	174.352	174.352	−482.480	0.000	0.000	0.000
	4700.00	33.472	143.509	105.701	177.699	177.699	−496.795	0.000	0.000	0.000
	4800.00	33.472	144.214	106.496	181.046	181.046	−511.182	0.000	0.000	0.000
	4871.00	33.472	144.706	107.050	183.423	183.423	−521.438	0.000	0.000	0.000

References

Phase	H / S	C_p	Remarks	
SOL − A	Hu1	Hu1		hcp
SOL − B	Hu1	Hu1		bcc
LIQ	Hu1	Hu1	Hu1	BPT = 4871., L = 575.46 kJ

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178.490 HAFNIUM (GAS) **Hf[g]**

Phase	T [K]	C_p	S	−(G−H298)/T	H	H−H298	G	ΔH_f	ΔG_f	log K_f [−]
		[J / (K mol)		]	[kJ / mol				]	
GAS	298.15	20.803	186.892	186.892	619.232	0.000	563.510	619.232	576.496	−101.000
	300.00	20.807	187.020	186.892	619.270	0.038	563.164	619.223	576.231	−100.331
	400.00	20.962	193.025	187.710	621.358	2.126	544.148	618.697	561.978	−73.387
	500.00	21.395	197.743	189.261	623.473	4.241	524.602	618.123	547.864	−57.235
	600.00	22.115	201.704	191.013	625.647	6.415	504.624	617.531	533.868	−46.477
	700.00	23.011	205.179	192.793	627.902	8.670	484.277	616.945	519.970	−38.801
	800.00	23.990	208.315	194.540	630.252	11.020	463.600	616.377	506.156	−33.049
	900.00	24.980	211.198	196.233	632.700	13.468	442.622	615.831	492.411	−28.579
	1000.00	25.924	213.879	197.865	635.246	16.014	421.367	615.307	478.726	−25.006
	1100.00	26.795	216.391	199.436	637.883	18.651	399.852	614.797	465.093	−22.085
	1200.00	27.598	218.758	200.949	640.603	21.371	378.094	614.295	451.506	−19.654
	1300.00	28.302	220.995	202.405	643.399	24.167	356.105	613.792	437.960	−17.597
	1400.00	28.900	223.115	203.810	646.260	27.028	333.899	613.278	424.454	−15.837
	1500.00	29.397	225.127	205.164	649.175	29.943	311.486	612.742	410.985	−14.312
	1600.00	29.800	227.037	206.472	652.136	32.904	288.877	612.175	397.553	−12.979
	1700.00	30.121	228.854	207.736	655.133	35.901	266.081	611.568	384.157	−11.804
	1800.00	30.371	230.583	208.957	658.158	38.926	243.109	610.914	370.799	−10.760
	1900.00	30.561	232.230	210.139	661.205	41.973	219.967	610.205	357.478	−9.828
	2000.00	30.702	233.802	211.283	664.268	45.036	196.665	609.436	344.197	−8.989
	2100.00	30.804	235.302	212.392	667.344	48.112	173.209	602.069	331.242	−8.239
	2200.00	30.878	236.737	213.466	670.428	51.196	149.607	601.471	318.359	−7.559
	2300.00	30.932	238.111	214.508	673.519	54.287	125.864	600.880	305.504	−6.938
	2400.00	30.976	239.428	215.519	676.614	57.382	101.987	600.294	292.674	−6.370
	2500.00	31.019	240.693	216.501	679.714	60.482	77.980	599.711	279.868	−5.848
	2600.00	31.069	241.911	217.455	682.818	63.586	53.850	575.410	268.042	−5.385
	2700.00	31.134	243.085	218.382	685.928	66.696	29.600	575.173	256.225	−4.957
	2800.00	31.222	244.218	219.285	689.046	69.814	5.234	574.944	244.416	−4.560
	2900.00	31.314	245.316	220.164	692.172	72.940	−19.243	574.723	232.615	−4.190
	3000.00	31.430	246.379	221.020	695.309	76.077	−43.828	574.513	220.822	−3.845
	3100.00	31.568	247.412	221.855	698.459	79.227	−68.518	574.315	209.036	−3.522
	3200.00	31.729	248.417	222.669	701.624	82.392	−93.309	574.133	197.255	−3.220
	3300.00	31.913	249.396	223.464	704.806	85.574	−118.200	573.967	185.481	−2.936
	3400.00	32.120	250.351	224.241	708.007	88.775	−143.188	573.822	173.710	−2.669
	3500.00	32.350	251.286	225.001	711.230	91.998	−168.270	573.698	161.944	−2.417
	3600.00	32.602	252.201	225.743	714.478	95.246	−193.444	573.598	150.181	−2.179
	3700.00	32.873	253.098	226.471	717.751	98.519	−218.709	573.524	138.421	−1.954
	3800.00	33.164	253.978	227.183	721.053	101.821	−244.063	573.479	126.662	−1.741
	3900.00	33.472	254.843	227.881	724.385	105.153	−269.504	573.463	114.904	−1.539
	4000.00	33.795	255.695	228.566	727.748	108.516	−295.031	573.479	103.146	−1.347
	4100.00	34.132	256.533	229.238	731.144	111.912	−320.643	573.529	91.387	−1.164
	4200.00	34.482	257.360	229.898	734.575	115.343	−346.338	573.612	79.627	−0.990
	4300.00	34.841	258.176	230.546	738.041	118.809	−372.115	573.731	67.864	−0.824
	4400.00	35.209	258.981	231.183	741.543	122.311	−397.973	573.886	56.098	−0.666
	4500.00	35.584	259.776	231.809	745.083	125.851	−423.910	574.078	44.328	−0.515
	4600.00	35.963	260.563	232.426	748.660	129.428	−449.928	574.309	32.553	−0.370
	4700.00	36.345	261.340	233.033	752.276	133.044	−476.023	574.577	20.773	−0.231
	4800.00	36.727	262.109	233.631	755.929	136.697	−502.195	574.883	8.986	−0.098
	4900.00	37.109	262.871	234.220	759.621	140.389	−528.444	0.000	0.000	0.000
	5000.00	37.488	263.624	234.800	763.351	144.119	−554.769	0.000	0.000	0.000

References

Phase	H / S	C_p
GAS	Hu1	Hu1

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HfB2 HAFNIUM DIBORIDE 200.112

Phase	T [K]	C_p	S	−(G−H298)/T	H	H−H298	G	ΔH_f	ΔG_f	log K_f
		[J / (K mol)]			[kJ / mol]					[−]
SOL	298.15	49.453	42.677	42.677	−335.975	0.000	−348.699	−335.975	−332.237	58.206
	300.00	49.748	42.984	42.678	−335.883	0.092	−348.778	−335.973	−332.213	57.843
	400.00	60.741	58.989	44.785	−330.293	5.682	−353.889	−335.726	−330.993	43.223
	500.00	66.818	73.249	49.083	−323.892	12.083	−360.516	−335.475	−329.841	34.458
	600.00	70.961	85.816	54.179	−316.992	18.983	−368.482	−335.302	−328.732	28.619
	700.00	74.189	97.006	59.512	−309.730	26.245	−377.634	−335.191	−327.646	24.449
	800.00	76.931	107.095	64.840	−302.171	33.804	−387.847	−335.116	−326.574	21.323
	900.00	79.389	116.301	70.054	−294.353	41.622	−399.023	−335.056	−325.509	18.892
	1000.00	81.672	124.784	75.108	−286.299	49.676	−411.083	−334.997	−324.452	16.948
	1100.00	83.839	132.671	79.987	−278.022	57.953	−423.960	−334.927	−323.400	15.357
	1200.00	85.929	140.056	84.688	−269.533	66.442	−437.601	−334.839	−322.356	14.032
	1300.00	87.963	147.015	89.217	−260.838	75.137	−451.957	−334.726	−321.320	12.911
	1400.00	89.958	153.607	93.583	−251.942	84.033	−466.991	−334.583	−320.294	11.950
	1500.00	91.922	159.880	97.795	−242.848	93.127	−482.668	−334.408	−319.280	11.118

References

Phase	H / S	C_p	Remarks	
SOL	Nb1	Nb1,e	Tk1	MPT= 3523.

HfBr4 HAFNIUM TETRABROMIDE 498.106

Phase	T [K]	C_p	S	−(G−H298)/T	H	H−H298	G	ΔH_f	ΔG_f	log K_f
		[J / (K mol)]			[kJ / mol]					[−]
SOL	298.15	127.612	238.488	238.488	−767.346	0.000	−838.451	−767.346	−734.703	128.717
	300.00	127.729	239.278	238.490	−767.110	0.236	−838.893	−767.437	−734.500	127.888
	400.00	134.072	276.892	243.576	−754.020	13.326	−864.776	−825.925	−711.313	92.888
	500.00	140.415	307.491	253.389	−740.295	27.051	−894.041	−822.274	−683.073	71.360
	600.00	146.758	333.652	264.636	−725.937	41.409	−926.128	−818.118	−655.614	57.076
	700.00	153.101	356.751	276.176	−710.944	56.402	−960.670	−813.437	−628.893	46.929

References

Phase	H / S	C_p
SOL	Pa2	Pa2

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498.106 HAFNIUM TETRABROMIDE (GAS) **HfBr4**[g]

Phase	T [K]	C_p	S	−(G−H298)/T	H	H−H298	G	ΔH_f	ΔG_f	log K_f
		[J / (K mol)		]	[kJ / mol				]	[−]
GAS	298.15	103.555	418.928	418.928	−656.888	0.000	−781.791	−656.888	−678.043	118.790
	300.00	103.609	419.569	418.930	−656.696	0.192	−782.567	−657.024	−678.173	118.080
	400.00	105.512	449.674	423.022	−646.227	10.661	−826.097	−718.132	−672.633	87.837
	500.00	106.403	473.325	430.802	−635.626	21.262	−872.289	−717.605	−661.321	69.088
	600.00	106.897	492.772	439.558	−624.959	31.929	−920.623	−717.140	−650.109	56.597
	700.00	107.203	509.275	448.368	−614.253	42.635	−970.746	−716.746	−638.969	47.680
	800.00	107.409	523.605	456.897	−603.522	53.366	−1022.405	−716.428	−627.881	40.996
	900.00	107.557	536.265	465.026	−592.773	64.115	−1075.411	−716.188	−616.828	35.800
	1000.00	107.669	547.603	472.726	−582.011	74.877	−1129.614	−716.028	−605.797	31.644

References

Phase	H / S	C_p
GAS	Pa2	Pa2

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